

THE QUANT HANDBOOK

Architecture, Mathematics & Operational Bible of the Brent Quant Engine

Version:	1.0 — Production Edition
System Name:	Brent Multi-Asset Quantitative Trading Engine
Target Asset Classes:	Crypto (Perpetual Futures), Forex Majors, Global Equities
Author / Stakeholder:	Egerton & Brent Website Ecosystem
Mathematical Models:	Hurst Exponent, Yang-Zhang Volatility, Half-Kelly Criterion
Defensive Core:	Synthetic Stops, -3% Daily Circuit Breaker, Liquidation Gate
Mobile Command:	Telegram 2-Way Encrypted Control Hub (@Eclat_quant_bot)
Date of Publication:	September 2026

CORE PRINCIPLE: CAPITAL PRESERVATION ABOVE ALL

"The elements of good trading are: (1) cutting losses, (2) cutting losses, and (3) cutting losses. If you can follow these three rules, you may have a chance." — **Ed Seykota**

Most retail traders approach markets asking: *'How much money can I make today?'*

Institutional quantitative firms ask: *'What is the maximum I can lose on this trade, and how do I survive a black swan event?'*

This fundamental paradigm separates the 95% of retail accounts that blow up within 90 days from the quantitative systems that compound wealth over decades. This engine is engineered from the first line of code with capital preservation as the primary objective, and alpha generation as the mathematical consequence of discipline.

CHAPTER 1: SYSTEM ARCHITECTURE & DATA PIPELINE

The Brent Quant Engine operates on a decoupled **6-tier modular pipeline**. No individual strategy or module is permitted to execute trades directly; every decision must traverse institutional gates before order generation.

Layer	Component Name	Role & Operational Responsibility
Layer 1	Market Data Router	Aggregates real-time feeds from CCXT (Crypto), Alpaca (Stocks), and MetaTrader 5 (Forex). Normalizes ticks into pre-allocated NumPy RingBuffers.
Layer 2	Intelligence & Macro Guards	Calculates market regimes (Hurst Exponent, Volatility), tracks Session liquidity (London/NY overlap), and scans economic calendars / news sentiment.
Layer 3	Strategy Ensemble & Scanner	Runs Trend Following (EMA Cross), Mean Reversion (Bollinger + RSI), and Breakout channels. Ranks pairs via Relative Strength and Currency Strength.
Layer 4	Institutional Risk Gate	Enforces 10 pre-trade criteria, Half-Kelly position sizing, dynamic leverage capping, liquidation price buffers, and the -3.0% daily circuit breaker.
Layer 5	Stealth Execution Engine	Maintains private Synthetic Stops in RAM (invisible to exchange books). Manages 2-stage exits: 50% TP1 scale-out + Chandelier trailing stop.
Layer 6	Telemetry & Mobile Hub	Streams audit logs to SQLite (WAL mode), powers the Terminal UI Dashboard, and handles 2-way remote commands via Telegram.

CHAPTER 2: MATHEMATICAL FOUNDATIONS OF OUR EDGE

Quantitative trading relies on statistical edges rather than opinions or patterns. Our bot incorporates three institutional quantitative models:

1. The Hurst Exponent (H) — Market Memory & Persistence

Developed by Harold Edwin Hurst, the Hurst Exponent determines whether an asset series behaves as a random walk, exhibits trending persistence, or demonstrates mean-reverting behavior.

Rescaled Range Analysis (R/S Formulation):

$E[R(n) / S(n)] = C * n^H \text{ as } n \rightarrow \text{infinity}$

$\log(R(n) / S(n)) = H * \log(n) + \log(C)$

Regime Classification Rules:

- **H > 0.55 (Persistent / Trending):** Past moves predict future moves. Bot unlocks **TrendFollowingStrategy**.
- **H < 0.45 (Anti-Persistent / Mean Reverting):** Price oscillates around mean. Bot unlocks **MeanReversionStrategy**.
- **0.45 <= H <= 0.55 (Geometric Brownian Motion):** Random walk noise. Bot suppresses signals to prevent chop.

2. Yang-Zhang Volatility Estimator

Standard deviation of close prices ignores overnight gap risk and intraday high/low excursions. The Yang-Zhang estimator is minimum-variance, unbiased, and independent of both opening jump and drift.

$$\text{sigma_YZ}^2 = \text{sigma_open}^2 + k * \text{sigma_close}^2 + (1 - k) * \text{sigma_RS}^2$$

$$\text{where } k = 0.34 / (1.34 + (n + 1) / (n - 1))$$

This high-fidelity volatility estimator continuously calibrates our Average True Range (ATR) dynamic stops, ensuring stops expand during volatile expansions and tighten during consolidations.

3. Half-Kelly Information Sizing

To maximize compound geometric wealth while completely eliminating the risk of ruin, position size is determined via the Kelly Criterion, scaled by 0.5x (Half-Kelly) to account for parameter uncertainty:

$$f^* = (p * (b + 1) - 1) / b$$

$$f_{\text{trade}} = \min(0.5 * f^*, \text{MaxRiskPct})$$

- **p** = Empirical win rate across the rolling 30 trades (computed live from SQLite).
- **b** = Empirical payout ratio (Average Profit / Average Loss).
- **Edge Collapse Safeguard:** If edge collapses ($f^* \leq 0$), sizing automatically drops to zero, preventing drawdowns.

CHAPTER 3: DEFENSIVE ARCHITECTURE & THE 10 RISK GATES

Every potential order generated by any strategy must pass through **10 sequential pre-trade gates** before entering the execution queue. If any single gate fails, the order is dropped.

Gate #	Gate Name	Condition & Capital Protection Logic
Gate 1	Actionable Direction	Enforces strict directionality (LONG or SHORT only; neutral signals discarded).
Gate 2	Confidence Floor	Rejects any signal with algorithmic confidence score < 25%.
Gate 3	Regime Filter	Blocks trades if regime is HIGH_VOL_CHAOS or incompatible with strategy.
Gate 4	Session Guard	Verifies market open, spread tightness, and avoids illiquid off-hours sessions.
Gate 5	Event Blackout	Halts trade entries 15 minutes prior to high-impact economic news releases.
Gate 6	News Panic Filter	Scans live headlines using VADER NLP; blocks buys during negative sentiment crashes.
Gate 7	Portfolio Capacity	Enforces max 10 concurrent positions and validates capital availability.
Gate 8	Volatility Sizing	Scales position size inversely to ATR; risk is fixed at exactly 1.0% of equity.
Gate 9	Broker Minimum	Validates order size against exchange minimum notional / lot size rules.
Gate 10	Liquidation Guard	(Leverage Gate) Rejects any trade where liquidation price is within 1.5x ATR of entry.

The -3.0% Daily Circuit Breaker

EMERGENCY PROTOCOL (DAILY CIRCUIT BREAKER):

If cumulative realized + unrealized daily losses reach **-3.0% of opening equity**:

1. The engine triggers an immediate **Kill Switch**.
2. All open positions are liquidated via aggressive IOC market orders across all brokers.
3. All pending synthetic stops and limit orders are revoked.
4. An emergency alert is dispatched to your Telegram phone.
5. All trading is strictly locked until the 00:00 UTC midnight reset.

CHAPTER 4: ALPHA ENGINES & PROFIT MAXIMIZATION

To maximize returns while locking in safety, the engine implements two proprietary alpha drivers:

1. Partial Scale-Out (TP1) & The "Free Ride" Chandelier Trail

Instead of closing 100% of a trade at a static take-profit or watching winning trades retrace to break-even, the execution engine runs an automated 3-stage exit:

Exit Stage	Trigger Price Coordinate	Execution Logic & Capital Outcome
Stage 1: TP1	Entry + (2.0 x ATR)	Instantly closes 50% of the position , banking guaranteed cash profit.
Stage 2: Risk-Free	Entry + (0.2 x ATR)	Concurrently raises the stop-loss on the remaining 50% past break-even. The trade is now mathematically risk-free.
Stage 3: Runner Trail	Highest High - (2.5 x ATR)	Chandelier trailing stop ratchets upwards with every new high, riding trends for days without capping upside.

2. Market Scanner: Relative Strength & Currency Strength Meter (CSM)

The engine does not cycle symbols alphabetically or pick trades at random. It ranks market breadth every 15 minutes:

- **Crypto & Equities (RS Ratio):** Ranks all assets by (Momentum / Volatility Ratio). Allocates capital exclusively to the top 3 highest-momentum leaders.
- **Forex (Currency Strength Meter):** Decomposes 28 forex pairs into net currency vectors for USD, EUR, GBP, JPY, AUD, CAD, CHF, NZD. Automatically pairs the **absolute strongest currency** against the **absolute weakest currency** for maximum directional impulse.

CHAPTER 5: STEALTH EXECUTION & PRIVATE STOPS

Retail traders consistently lose money because public exchange order books broadcast stop-loss orders to predatory High-Frequency Trading (HFT) algorithms and market maker liquidity sweeps.

THE SYNTHETIC STOP PROTOCOL:

1. **Zero Footprint:** Stop-loss and take-profit coordinates exist **only in private server RAM**. No resting orders are ever placed on exchange order books.
2. **Millisecond Tick Monitor:** The engine evaluates live bid/ask prices against private thresholds on every tick.
3. **Aggressive IOC Execution:** When a stop is breached, an Immediate-Or-Cancel market order is dispatched instantly.
4. **Institutional Stealth:** Market makers cannot see or front-run our stops because the order does not exist in the book until the millisecond it is executed.

CHAPTER 6: 2-WAY MOBILE COMMAND CENTER

The engine connects directly to your personal phone via @Eclat_quant_bot. Incoming requests are filtered through your strict Telegram User ID, blocking all unauthorized traffic.

Command	Function & Return Telemetry
/status	Returns current portfolio equity, daily P&L; (\$ and %), circuit breaker status, and active position count.
/positions	Inspects all open trades: symbol, entry, mark price, P&L;, stop price, and [TRAIL] / [BE] indicators.
/pause	Suspends new trade entries ahead of uncertain news while keeping trailing stops active on existing trades.
/resume	Restores automatic regime scanning, market evaluation, and trade entry execution.

/closeall	Emergency Panic Button: Instantly market-closes 100% of open positions across all asset classes.
/report	Generates a performance summary of today's closed trades, win rate, and net P&L; from SQLite audit logs.
/help	Displays mobile command directory and operational syntax.

CHAPTER 7: OPERATIONAL PLAYBOOK & COMMANDS

```
# 1. Starting the Bot (Paper Mode):
cd "c:\Users\egerton\Desktop\BRENT WEBSITE\quant_engine"
python main.py --mode paper

# 2. Running the 32-Test Verification Suite:
python -m pytest tests/ -v --tb=short

# 3. Previewing the Terminal Dashboard Alone:
python -m monitoring.dashboard
```

CHAPTER 8: GROWTH & RESEARCH ROADMAP

As capital compounds and the account matures, the engine follows an institutional milestone expansion roadmap:

Phase	Milestone Name	Status	Deliverable Scope
Phase 1	Core Engine + Tests	COMPLETE	Multi-asset router, regime filter, ring buffers, 20 tests.
Phase 2	Live Terminal UI	COMPLETE	6-panel Rich UI dashboard with live position cards.
Phase 3	Safe Leverage Mode	COMPLETE	Liquidation buffer checks, leverage tier manager.
Phase 4	Alpha Expansion	COMPLETE	Chandelier trailing stop, Relative Strength, Half-Kelly.
Phase 5	2-Way Mobile Hub	COMPLETE	Telegram bot command polling & alerts (@Eclat_quant_bot).
Phase 6	FastAPI Web UI	UPCOMING	Modern web dashboard with TradingView candlestick charts.
Phase 7	Automated Profit Vault	UPCOMING	Sweeps 30% of weekly profits into cold reserve wallet.
Phase 8	Cloud VPS 24/7	UPCOMING	Deploy systemd service on Linux VPS for zero-downtime execution.
Phase 9	Walk-Forward Optimizer	UPCOMING	Auto-calibrates EMA lengths and stop multipliers quarterly.

"The Quant Handbook is maintained under active quantitative governance. Every modification to strategies, risk multipliers, and execution models should be documented within these pages as our system compounds."